CS 215: Task 08: Due: 30-Oct; time 7:00 AM

- Your answers should be in your handwriting. Try to be neat and clear. If an LLM can read your handwriting, we might be able to.
- If you used any source (person or thing) explicitly state it. This is an individual task.

Teaching-team notes.

- For all questions in this task, explicitly write down the assumptions made.
- For all problems here, do not use electronic calculating aids since these will not be allowed in the exam. You may refer to the tables mentioned in the appendix of the textbook (5th edition).

---

0. Please write the honor code in your handwriting, with the understanding that it is true.

1. In class, we derived the maximum likelihood estimator (MLE) for a normal distribution by first writing the probability density for a single observation x_i, then constructing the likelihood of the entire dataset using independence, and finally maximizing the log-likelihood.

   We now apply the same approach to a different probability model.

   Teaching-team notes. First, review the example in the book on the Bernoulli case. We are going from scalars to vectors in this question.

   In a population, a gene has two alleles, A [dominant] and a [recessive]. Every indvidual gets two copies of this gene (one from each parent) so there are three possible combinations (genotypes): AA, Aa, and aa. It is easy to see that they occur with probabilities

   $$p^2, \qquad 2p(1-p), \qquad (1-p)^2,$$

   respectively, where p is the unknown population frequency of the allele A.

   Suppose we independently sample N individuals. We observe counts of n_1 for AA, n_2 for Aa, and n_3 for aa (where $n_1 + n_2 + n_3 = N$).

   For each individual i, represent the observed genotype by a one-hot vector

   $$x_i = (x_{i1}, x_{i2}, x_{i3}),$$

   where $x_i = (1, 0, 0)$ denotes AA, $x_i = (0, 1, 0)$ denotes Aa, and $x_i = (0, 0, 1)$ denotes aa.

   (a) Write an expression for the probability $P(X_i = x_i; p)$ of a single observation, in terms of the components of x_i and the parameter p.

(b) Using the independence of the observations, construct the likelihood function $L(p)$ for the entire dataset $x_1, \ldots, x_N$.

(c) Derive the maximum likelihood estimator $\hat{p}$. Express your answer in terms of the numbers n_1, n_2, n_3 of individuals with genotypes AA, Aa, aa, respectively.

2. Let $X_1, \ldots, X_n$ be independent observations from a uniform distribution $U(0, \theta)$, where $\theta > 0$ is unknown.

(a) Write the likelihood function $L(\theta)$, explicitly stating the values of θ for which it is nonzero.

(b) Derive the maximum likelihood estimator of θ.

(c) Is the estimator unbiased? Justify your answer.

3. The returns on financial assets for (volatile) cryptocurrency are sometimes modeled using a Laplace distribution, which has heavier tails (extreme jumps are more common) than a Normal distribution.

   Suppose $X_1, \ldots, X_n$ are independent observations from a Laplace distribution with probability density function

$$f(x; \mu, b) = \frac{1}{2b} \exp\left(-\frac{|x - \mu|}{b}\right),$$

   where μ is an unknown location parameter and $b > 0$ is a known scale parameter.

   (a) Write down the likelihood and log-likelihood functions for the observed data $x_1, \ldots, x_n$.

   (b) Derive the maximum likelihood estimator of μ.

   (c) Compare this estimator with the MLE of the mean of a Normal distribution. Which estimator is less sensitive to extreme observations, and why?

4. A sample of 20 cigarettes is tested to determine nicotine content, and the average value observed is 1.2 mg. Suppose that the population variance is not known in advance, and the sample variance is 0.04.

   Assuming a normal population, compute a 99% two-sided confidence interval for the mean nicotine content of a cigarette.

5. A random sample of 1200 engineers includes 300 Hispanic American engineers, 400 African American engineers. Assume no overlap.

   Determine 90% confidence intervals

   (a) For the proportion of all engineers who are Hispanic American or African American.

   (b) The excess in the proportion of African Americans and proportion of Hispanic Americans

6. It is claimed that a certain type of bipolar transistor has a mean current gain of at least 210. A sample of these transistors has sample mean current gain 200 and sample standard deviation 35.

   Assuming that the population is normal, determine whether the claim should be rejected at the 5% level of significance if

   (a) the sample size is 25;

   (b) the sample size is 64.